Krish Puwar

+1 (978) 288-6811 | krishpuwar.jobs@gmail.com | krishpuwar.netlify.app

Education

The State University of New York at Buffalo — **Buffalo, NY**
Master of Science in Computer Science - Computer Security and Privacy Specialization — *August 2026 – December 2027*

The State University of New York at Buffalo — **Buffalo, NY**
Bachelor of Science in Computer Engineering — Cum Laude, GPA 3.44 — Dean's List — *May 2026*

Experience

Graduate Research Assistant — **September 2026 – Present**
University at Buffalo, advised by Prof. Hongxin Hu — *Buffalo, NY*

- Contributing to **hands-on AI security lab development** focused on secure coding and the security evaluation of LLM-generated code.
- Analyzing research benchmarks (**CWEval, BaxBench, DualGauge**) and exploring **CodeQL and Semgrep** for static analysis and vulnerability detection.

Undergraduate Research Engineer Intern — **February 2026 – May 2026**
Integrated Systems — *Victor, NY*

- Contributing to **GPROP**, an RF signal propagation modeling tool built in the WN4SS Lab as part of an **SPIR -funded** industry collaboration simulating multipath propagation in IoT-dense urban environments.
- Authored **Bash automation scripts** chaining GPROP's fetch, preprocessing, and model execution into one pipeline.
- Built **containerized Docker environments** standardizing the pipeline across dev and deployment targets.
- Documented system architecture, pipeline design, and experimental results for a forthcoming **research publication**.

Security Engineering Intern — **July 2025 – August 2025**
Zayed University — *Abu Dhabi, UAE*

- Worked at the university's on-campus **Security Operations Center (SOC)**, engineering a full-stack SOC training platform (**Flask, PostgreSQL, Docker**) with a normalized schema and REST API contracts for real-time performance analytics.
- Used **MITRE Caldera** for **red-team attack simulation**, chaining ATT&CK-mapped TTPs to emulate real adversary behavior, paired with **Wazuh** monitoring to drive live detection-and-response exercises.
- Built **AI-driven automation** pipelines with the Gemini API to generate scenario content and score exercises dynamically, reducing manual SOC training overhead.
- Deployed and maintained the platform via **Docker** on university infrastructure, ensuring dev/prod environment parity for scalable multi-user training operations.

Projects

AWS SOC SIEM | *AWS (EC2, S3, Lambda,SAM), Splunk, MITRE ATT&CK* | GitHub — June 2026 – August 2026

- Built a **cloud-native SIEM** that ingests security logs into S3, parses and scans them through Lambda functions running **5 MITRE ATT&CK-mapped detection rules**, and stores structured alerts in DynamoDB.
- Engineered a **threat-intelligence enrichment service** integrating the VirusTotal and AbuseIPDB APIs to automatically score detected indicators of compromise (IPs, hashes, domains), flag malicious activity, and block offending IPs via EC2.
- Deployed a self-hosted **Splunk Enterprise** instance on **EC2** and streamed alerts into it through the **HTTP Event Collector (HEC)**, replacing a custom dashboard with Splunk dashboards that map detections to MITRE ATT&CK tactics and techniques for analyst triage.
- Defined the full stack as **infrastructure-as-code** via an AWS SAM template with **least-privilege IAM**, enabling reproducible, version-controlled deployments across environments.

SQL Injection Attack & Defense Lab | *PHP, MySQL, Docker, cURL* | GitHub — August 2026 - October 2026

- Deployed a containerized **SEED Labs** environment on a Ubuntu VM (ARM64), configuring **Docker Compose** multi-container networking (web server + MySQL) and resolving LVM storage constraints to support the lab environment.
- Exploited an **authentication bypass** vulnerability in a PHP/MySQL login system using crafted SQL injection payloads (`' #`), retrieving all employee records without valid credentials.
- Escalated privileges via **UPDATE-statement injection**, hijacking `WHERE` clauses to modify arbitrary users' salary and password fields from a low-privilege session, including injecting a precomputed **SHA1** hash to achieve full account takeover.
- Fixed the **SQL injection** vulnerability on the web server by implementing **prepared statements**, verifying that previously successful injection payloads were neutralized.

Technical Skills

Languages: Python, SQL, Bash, C, Java
Security: SIEM (Wazuh), Threat Intelligence, MITRE ATT&CK, Adversary Emulation (Caldera), Cryptography, OpenSSL
Cloud: AWS (EC2, S3, Lambda, DynamoDB, IAM, SAM), Infrastructure as Code, CloudFormation
Virtualization: Nutanix (Cloud Infrastructure, AHV, Cloud Manager, Kubernetes Platform), Hyper-V, vSphere
Networking: TCP/IP, Network Protocols, iptables, Firewall Configuration, Mininet, Log Analysis, Security Hardening
Data & Databases: PostgreSQL, CodeQL, Pandas, NumPy, Splunk, Tableau, Power BI, Query Optimization, Indexing
Systems & DevOps: Linux, Bash Scripting, Python Scripting, Docker, Git
Certifications: CompTIA Security+ (SY0-701) *Expected 2026* | CCNA *Expected 2026* | AWS CLF 002 *October 2026*